	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE **Supplementary Examination – Summer 2022** **Course: B. Tech.SE Branch :CSE Semester :-III** **Subject Code & Name:Computer Architecture & Organization (BTCOC304)** **Max Marks: 60 Date: Duration: 3 Hr.**		
	Instructions to the Students: 1. *All the questions are compulsory.* 2. *The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.* 3. *Use of non-programmable scientific calculators is allowed.* 4. *Assume suitable data wherever necessary and mention it clearly.*		
		(Level/CO)	Marks
Q. 1	**Solve Any Two of the following.** *(This is just a sample instruction)*		
A)	**Write a note on ALU design.**		**06**
B)	**Write a note on computer components.**		**06**
C)	**Define terms: Single-cycle and multi-cycle Implementation**		**06**
Q.2	**Solve Any Two of the following.** *(This is just a sample instruction)*		
A)	**Explain RISC and CISC architecture.**		**06**
B)	**Draw and explain single-cycle datapath for the memory instructions,**		**06**
C)	**Explain in detail elements of Bus design.**		**06**
Q. 3	**Solve Any One of the following.** *(This is just a sample instruction)*		
A)	**Explain the following addressing modes: i]Immediate ii]Indirect iii]Displacement**		**06**
B)	**Write a short note on Disk Storage.**		**06**
C)	**Draw and explain in detail the concept of virtual memory with page fault.**		**06**
Q.4	**Solve Any Two of the following.** *(This is just a sample instruction)*		
A)	**Draw and explain RAID levels in detail.**		**006**
B)	**Write a note on the Hardwired Control unit.**		**06**
C)	**Write a note on Magnetic tape and Optical Memory.**		**06**
Q. 5	**Solve Any One of the following.** *(This is just a sample instruction)*		
A)	**Define and explain in detail Data Hazard with examples.**		**06**

B)	**Draw and explain in detail Memory Hierarchy. Difference between main memory and cache memory.**		**06**
C)	**What is DMA?Draw typical DMA module diagram.**		**06**
	***** End *****		

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